

# Noise-Floor Estimation for FMCW Radar Range–Doppler Segments: Derivation from the Exponential Distribution of Circular Complex Gaussian Noise

fmwc\_radar project

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## Abstract

This note derives, from first principles, the noise-power (noise-floor) estimator used in the SNR-normalisation pipeline of the `fmwc_radar` project (`snr_norm.py`). The estimator recovers the noise variance  $\sigma^2$  of circularly-symmetric complex Gaussian thermal noise from a single low percentile of the measured power in azimuth bins known to lie outside the radar’s main beam. The key fact exploited is that the squared magnitude of complex Gaussian noise is exponentially distributed, which gives a closed-form map from any percentile of the power distribution back to  $\sigma^2$ . We also derive the companion noise-injection formula used to force a segment to a target SNR.

## 1 Signal model

Let  $z \in \mathbb{C}$  denote one complex (I/Q) sample of the receiver output in a range–azimuth (“range–Doppler” before the slow-time FFT) bin that contains *no* target return — e.g. an azimuth column far outside the antenna’s main beam. Thermal receiver noise is modelled as circularly-symmetric complex additive white Gaussian noise (CSCG):

$$z = n_I + j n_Q, \quad n_I, n_Q \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{\sigma^2}{2}\right), \quad (1)$$

so that  $\mathbb{E}[z] = 0$ ,  $\text{Var}(n_I) = \text{Var}(n_Q) = \sigma^2/2$ , and the total noise power is

$$\mathbb{E}[|z|^2] = \mathbb{E}[n_I^2] + \mathbb{E}[n_Q^2] = \sigma^2. \quad (2)$$

$\sigma^2$  is exactly the quantity the estimator must recover: the noise floor in linear power units.

## 2 Why $|z|^2$ is exponentially distributed

Define  $P = |z|^2 = n_I^2 + n_Q^2$ . Since  $n_I, n_Q$  are independent zero-mean Gaussians with equal variance  $\sigma^2/2$ , the pair  $(n_I, n_Q)$  is a 2-D isotropic Gaussian vector. Converting to polar coordinates  $n_I = r \cos \theta$ ,  $n_Q = r \sin \theta$ , the joint density is

$$f(n_I, n_Q) = \frac{1}{\pi \sigma^2} \exp\left(-\frac{n_I^2 + n_Q^2}{\sigma^2}\right) = \frac{1}{\pi \sigma^2} \exp\left(-\frac{r^2}{\sigma^2}\right). \quad (3)$$

Because this depends only on  $r^2 = P$ , the phase  $\theta$  is uniform on  $[0, 2\pi)$  and independent of  $P$  (this is precisely what “circularly symmetric” means). Changing variables from  $(r, \theta)$  to  $(P, \theta)$  with  $P = r^2$ ,  $dP = 2r dr$ , the density of  $P$  alone is obtained by integrating out  $\theta$ :

$$f_P(p) = \int_0^{2\pi} \frac{1}{\pi\sigma^2} e^{-p/\sigma^2} \cdot \frac{1}{2} d\theta = \frac{1}{\sigma^2} e^{-p/\sigma^2}, \quad p \geq 0. \quad (4)$$

Equation (4) is the density of an  $\text{Exponential}(\lambda)$  random variable with rate  $\lambda = 1/\sigma^2$ , i.e.

$$P = |z|^2 \sim \text{Exponential}\left(\frac{1}{\sigma^2}\right), \quad \mathbb{E}[P] = \sigma^2, \quad \text{Var}(P) = \sigma^4. \quad (5)$$

This is the standard result that the squared envelope of narrowband Gaussian noise (equivalently, the noise power in a single range-Doppler bin) follows an exponential law, while the envelope  $r = |z|$  itself follows a Rayleigh distribution — the two are related by  $P = r^2$ .

### 3 Percentile-to- $\sigma^2$ inversion

The exponential CDF is

$$F_P(p) = \Pr(P \leq p) = 1 - e^{-p/\sigma^2}. \quad (6)$$

Let  $q_\alpha$  denote the  $\alpha$ -th percentile of  $P$  ( $\alpha \in (0, 100)$ ), i.e. the value solving  $F_P(q_\alpha) = \alpha/100$ . Solving (6) for  $q_\alpha$ :

$$1 - e^{-q_\alpha/\sigma^2} = \frac{\alpha}{100} \implies e^{-q_\alpha/\sigma^2} = 1 - \frac{\alpha}{100} \implies q_\alpha = -\sigma^2 \ln\left(1 - \frac{\alpha}{100}\right). \quad (7)$$

Inverting (7) gives the estimator actually implemented in code:

$$\boxed{\hat{\sigma}^2 = \frac{q_\alpha}{-\ln(1 - \alpha/100)}}, \quad c(\alpha) \equiv -\ln\left(1 - \frac{\alpha}{100}\right). \quad (8)$$

In `full_edge_sigma2()` this is written literally as

```
corr = -np.log(1.0 - percentile / 100.0)
sigma2 = np.percentile(p_edge, percentile) / corr
```

where `p_edge` is the empirical sample of  $|z|^2$  pooled from the off-beam azimuth columns (“far edges” of the 256-sample sweep) across all 5 range rows.

#### 3.1 Special case: the median

For  $\alpha = 50$  (the median),  $c(50) = -\ln(0.5) = \ln 2$ , so

$$\hat{\sigma}^2|_{\alpha=50} = \frac{\text{median}(P)}{\ln 2} \approx \frac{\text{median}(P)}{0.6931}. \quad (9)$$

This is the identity used by `snr_peak_floor`’s crude floor estimate, and is the textbook relation between the median and the mean of an exponential distribution (median =  $\sigma^2 \ln 2$ , since  $\mathbb{E}[P] = \sigma^2$ ).

### 3.2 Why a low percentile ( $\alpha = 10$ ), not the median, is used

The off-beam columns are only *approximately* pure noise: sidelobe leakage from a strong in-beam target can only ever push  $|z|^2$  *upward*, never downward, relative to the true noise floor. The exponential model is therefore a good fit for the *lower* tail of the empirical distribution and progressively contaminated in the upper tail. Because  $c(\alpha) = -\ln(1 - \alpha/100)$  is monotonically increasing in  $\alpha$  (e.g.  $c(10) = 0.105$ ,  $c(25) = 0.288$ ,  $c(50) = 0.693$ ,  $c(90) = 2.303$ ), a small positive bias  $\delta$  added to the raw percentile  $q_\alpha$  is divided by a *smaller* constant at low  $\alpha$ , i.e. contamination is attenuated less in relative terms — the practical benefit of  $\alpha = 10$  over  $\alpha = 50$  is not that the formula suppresses contamination algebraically, but that a low percentile of the empirical order statistics sits in a region of the distribution that leakage has not yet reached, giving an estimate closer to the true, uncontaminated  $\sigma^2$ . This was confirmed empirically in `snr_percentile_sweep_diagnostic.py`: lowering  $\alpha$  from 50 to  $\approx 10$ –25 roughly halved the excess bias observed for the corner-reflector class ( $\text{RCS} = 100 \text{ m}^2$ ) relative to the other eight classes, at the cost of a modest increase in per-estimate variance (fewer effective order-statistic samples are averaged near the extreme tail).

### 3.3 Bias–variance trade-off in $\alpha$

From (5), the  $\alpha$ -th sample percentile of  $n$  iid exponential draws is asymptotically normal,

$$\hat{q}_\alpha \sim \mathcal{N}\left(q_\alpha, \frac{\alpha(100 - \alpha)}{n f_P(q_\alpha)^2 100^2}\right), \quad (10)$$

by the standard delta-method result for sample quantiles, where  $f_P(q_\alpha) = \frac{1 - \alpha/100}{\sigma^2}$  from (4)–(6). Substituting and propagating through (8) shows the coefficient of variation of  $\hat{\sigma}^2$  grows as  $\alpha \rightarrow 0$  (few samples fall below a very low percentile) and as  $\alpha \rightarrow 100$  (the tail is sparse and, in this application, contaminated). The default  $\alpha = 10$  with  $n = 400$  pooled samples (5 range rows  $\times$  80 edge columns) is chosen empirically to sit in the favourable region of this trade-off.

## 4 From noise floor to SNR

Let  $P_{\text{peak}}$  be the measured power in the main-beam peak window on the centre range row (the paper’s convention: “stated SNRs always refer to the SNR at the centre of  $s_3$ ”). With  $\hat{\sigma}^2$  the noise floor from (8), the estimated signal power and linear SNR are

$$\hat{P}_{\text{sig}} = \max(P_{\text{peak}} - \hat{\sigma}^2, 0), \quad (11)$$

$$\hat{\gamma} = \frac{\hat{P}_{\text{sig}}}{\hat{\sigma}^2}, \quad \widehat{\text{SNR}}_{\text{dB}} = 10 \log_{10} \hat{\gamma}. \quad (12)$$

This is exactly `snr_full_edge(): signal = max(p_c - sigma2, 0), return 10*log10(signal/sigma2)`.

## 5 Noise injection to a target SNR

To force a segment currently at SNR  $\gamma_{\text{cur}} = 10^{\text{SNR}_{\text{cur}}/10}$  down to a target  $\gamma_t = 10^{\text{SNR}_t/10}$  ( $\gamma_t < \gamma_{\text{cur}}$ ), write the current peak power as  $P_{\text{sig}} = \hat{\sigma}^2 \gamma_{\text{cur}}$  (from the definition of  $\hat{\gamma}$  above; the signal power itself is not touched by adding independent noise). After adding independent CSCG noise of variance  $\Delta$ , the new floor is  $\hat{\sigma}^2 + \Delta$  and the new SNR is

$$\gamma_t = \frac{P_{\text{sig}}}{\hat{\sigma}^2 + \Delta} = \frac{\hat{\sigma}^2 \gamma_{\text{cur}}}{\hat{\sigma}^2 + \Delta}. \quad (13)$$

Solving for  $\Delta$ :

$$\boxed{\Delta = \hat{\sigma}^2 \left( \frac{\gamma_{\text{cur}}}{\gamma_t} - 1 \right)}, \quad \Delta \geq 0 \text{ enforced (noise can only be added, never removed).} \quad (14)$$

This matches `add_noise_to_target()`: `add_var = sigma2 * (g_cur/g_t - 1)`. The actual complex noise sample added per I/Q pair is drawn as

$$w = w_I + j w_Q, \quad w_I, w_Q \stackrel{\text{iid}}{\sim} \mathcal{N}\left(0, \frac{\Delta}{2}\right) \quad (15)$$

so that  $\mathbb{E}[|w|^2] = \Delta$ , consistent with the CSCG model of (1)–(2) and implemented as `sqrt(add_var/2) * (randn +`

## 6 Summary of the estimator chain

- (1) Off-beam power samples  $P_i = |z_i|^2$  are, under the CSCG noise model, iid Exponential( $1/\sigma^2$ ) (Eq. (5)).
- (2) Any percentile  $\alpha$  of an exponential distribution is an affine function of  $\sigma^2$  (Eq. (7)), so  $\sigma^2$  is recovered in closed form from a single empirical percentile via Eq. (8).
- (3) A low percentile ( $\alpha = 10$ ) is preferred over the median ( $\alpha = 50$ ) because target sidelobe leakage only inflates the upper tail of the empirical distribution; the low tail remains closer to the pure-noise exponential law.
- (4) The recovered  $\hat{\sigma}^2$  is used both to estimate the current SNR (Eq. (12)) and to size the exact variance of injected noise needed to reach a target SNR (Eq. (14)) — the same floor definition must be used in both steps, or the two computations become mutually inconsistent (a bug noted and fixed in the project history, see `snr_norm.py` docstring).